DeCovarT: A Probabilistic and Multidimensional Framework for Cellular Deconvolution in Heterogeneous Biological Samples

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Abstract

Although bulk transcriptomic analyses have greatly contributed to a better understanding of complex diseases, their sensibility is hampered by the highly heterogeneous cellular compositions of biological samples. To address this limitation, computational deconvolution methods have been designed to automatically estimate the frequencies of the cellular components that make up tissues, typically using reference samples of physically purified populations. However, they perform badly at differentiating closely related cell populations.

We hypothesised that the integration of the covariance matrices of the reference samples could improve the performance of deconvolution algorithms. We therefore developed a new tool, DeCovarT, that integrates the structure of individual cellular transcriptomic network to reconstruct the bulk profile. Specifically, we inferred the ratios of the mixture components by a standard maximum likelihood estimation (MLE) method, using the Levenberg-Marquardt algorithm to recover the maximum from the parametric convolutional distribution of our model. We then consider a reparametrisation of the log-likelihood to explicitly incorporate the simplex constraint on the ratios. Preliminary numerical simulations suggest that this new algorithm outperforms previously published methods, particularly when individual cellular transcriptomic profiles strongly overlap.
Purpose: The abstract serves both as a general introduction to the topic and as a brief, non-technical summary of the main results and their implications. The abstract must not include subheadings (unless expressly permitted in the journal's Instructions to Authors), equations or citations. As a guide the abstract should not exceed 200 words. Most journals do not set a hard limit however authors are advised to check the author instructions for the journal they are submitting to.
Methods: The abstract serves both as a general introduction to the topic and as a brief, non-technical summary of the main results and their implications. The abstract must not include subheadings (unless expressly permitted in the journal's Instructions to Authors), equations or citations. As a guide the abstract should not exceed 200 words. Most journals do not set a hard limit however, authors are advised to check the author instructions for the journal they are submitting to.
Results: The abstract serves both as a general introduction to the topic and as a brief, non-technical summary of the main results and their implications. The abstract must not include subheadings (unless expressly permitted in the journal's Instructions to Authors), equations or citations. As a guide the abstract should not exceed 200 words. Most journals do not set a hard limit however authors are advised to check the author instructions for the journal they are submitting to.
Conclusion: The abstract serves both as a general introduction to the topic and as a brief, non-technical summary of the main results and their implications. The abstract must not include subheadings (unless expressly permitted in the journal's Instructions to Authors), equations or citations. As a guide the abstract should not exceed 200 words. Most journals do not set a hard limit however authors are advised to check the author instructions for the journal they are submitting to.

Keywords: deconvolution, bulk RNASeq, Gaussian, generative model

1 Definitions

- The spectrum is the set of eigenvalues of a matrix.
- The Diag operator keeps only the diagonal elements of a matrix, setting off-diagonal terms as null.

- the condition number of a matrix $\boldsymbol{X}$ is given by the ratio of the largest eigenvalue over the smallest eigenvalue $\kappa(\boldsymbol{X}) = \frac{\lambda_{\max}}{\lambda_{\min}}$. A large condition number is a cue of strong multicollinearity (in other words, strong interactions between variables), leading to more complex inversion of matrices and loss of numerical precision.

2 Introduction

The analysis of the bulk transcriptome provided new insights on the mechanisms underlying disease development. However, such methods ignore the intrinsic cellular heterogeneity of complex biological samples, by averaging measurements over several distinct cell populations. Failure to account for changes of the cell composition is likely to result in a loss of *specificity* (genes mistakenly identified as differentially expressed, while they only reflect an increase in the cell population naturally producing them) and *sensibility* (genes expressed by minor cell populations are amenable being masked by highly variable expression from major cell populations). Accordingly, a range of computational methods have been developed to estimate cellular fractions, but they perform poorly in discriminating cell types displaying high phenotypic proximity. Indeed, most of them assume that purified cell expression profiles are fixed observations, omitting the variability and intrinsically interconnected structure of the transcriptome. For instance, the gold-standard deconvolution algorithm *CIBERSORT* [1] applies nu-support vector regression (ν-SVR) to recover the minimal subset of the most informative genes in the purified signature matrix. However, this machine learning approach assumes that the transcriptomic expressions are independent.

In contrast to these approaches, we hypothesised that integrating the pairwise covariance of the genes into the reference transcriptome profiles could enhance the performance of transcriptomic deconvolution methods. The generative probabilistic model of our algorithm, *DeCovarT* (Deconvolution using the Transcriptomic Covariance), is, to our knowledge, the first implementation of a *multivariate* Gaussian convolution for bulk deconvolution, with a sparse cell-type-specific covariance (in contrast to the univariate, gene-wise Gaussians of DSection [2]).

A key practical challenge for *DeCovarT* is that, unlike methods relying solely on mean expression profiles, it requires a covariance (or precision) matrix encoding gene–gene interactions. Recovering this structure from observational expression data alone is inherently difficult: expression correlations are largely correlative, and inferring directed, mechanistically interpretable regulatory edges typically requires additional data modalities such as chromatin accessibility (ATAC-seq) or transcription-factor binding information [3, 4]. Systematic benchmarking further shows that even with richer multi-omic inputs, GRN inference can fail in common regimes [5]. Whether recovering the true causal regulatory architecture is strictly necessary for accurate deconvolution, or whether statistically consistent covariance estimation is sufficient to improve compositional recovery, remains an open question that we address experimentally in the Results section.

3 Results

See online methods for details.

3.1 Methods overview

General explanation of DecovarT's optimisation framework

3.2 Numerical Simulation Study

3.2.1 Toy Model with two genes, and two cell populations

To assert numerically the relevance of accounting for the correlation between expressed transcripts, we designed a simple toy example with two genes and two cell proportions. Hence, using the simplex constraint (3), we only have to estimate one free unconstrained parameter, ρ_1, and then uses the mapping function, defined in Appendix A.3 to recover the ratios in their original space.

We simulated the bulk mixture, $\boldsymbol{y} \in \mathcal{M}_{G\times N}$, for a set of artificial samples $N = 500$, with the following generative model:

- We have tested two levels of cellular ratios, one with equi-balanced proportions $(\boldsymbol{p} = (p_1, p_2 = 1 - p_1) = (\frac{1}{2}, \frac{1}{2})$ and one with highly unbalanced cell populations: $\boldsymbol{p} = (0.95, 0.05)$.
- Then, each purified transcriptomic profile is drawn from a multivariate Gaussian distribution. We compared two scenarios, playing on the mean distance of centroids, respectively $\mu_{.1} = (20, 22), \mu_{.2} = (22, 20)$ and $\mu_{.2} = (20, 40), \mu_{.2} = (40, 20))$ and building the covariance matrix, $\boldsymbol{\Sigma} \in \mathcal{M}_{2\times 2}$ by assuming equal individual variances for each gene (the diagonal terms of the covariance matrix, $\operatorname{Diag}(\boldsymbol{\Sigma_1}) = \operatorname{Diag}(\boldsymbol{\Sigma_1}) = \boldsymbol{I}_2)$ but varying the pairwise correlation between gene 1 and gene 2, $\mathbb{C}\mathrm{ov}\,[x_{1,2}]$, on the following set of values: $\{-0.8, -0.6, \dots, 0.8\}$ for each of the cell population.
- As stated in 2, we assume that the bulk mixture, $\mathbf{y}_{.i}$ could be directly reconstructed by summing up the individual cellular contributions weighted by their abundance, without additional noise.

We compared the performance of DeCovarT algorithm with DeconRNASeq algorithm ([6]).

Even with a limited toy example including two cell populations characterised only by two genes, we observe that the overlap was a good proxy of the quality of the estimation: the less the overlap between the two cell distributions, the better the quality of the estimation 1.

To evaluate the biological and statistical interest of `DeCovarT`, we then expand the simulation framework, by encompassing a larger number of cell types, genes, and testing the sensitivity of the model by voluntarily including noise in the benchmark evaluation.

3.2.2 Scaling numerical simulations using methods for generating synthetic networks

The general framework for generating realistic simulations of single-cell profiles while modelling interactions between genes comprises the following steps. First, generate the

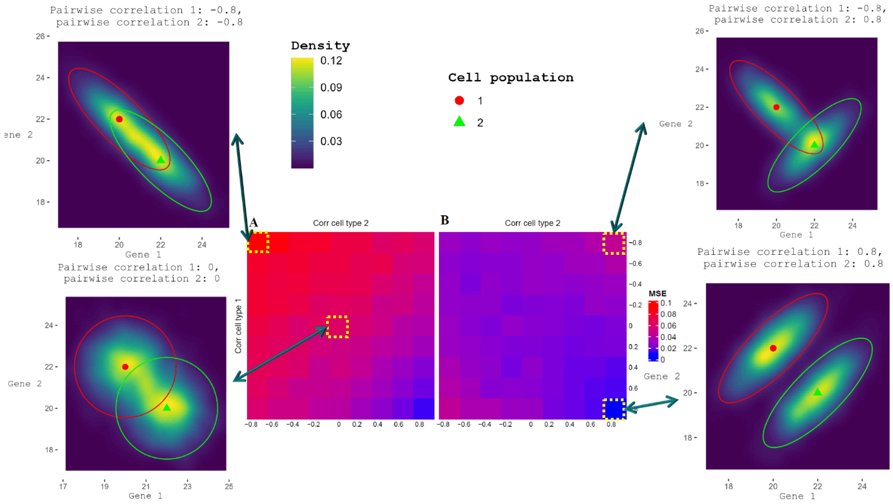

Fig. 1 We used the package **ComplexHeatmap** to display the mean square error (MSE) of the estimated cell ratios, comparing the NNLS output, as implemented in the DeconRNASeq algorithm ([6]), in Panel **A**, with our newly implemented DeCovarT algorithm, in Panel **B**. The lower the MSE, the least noisy and biased the estimates. In addition, we added the two-dimensional density plot for the intermediate scenario, for which each population is parameterised by a diagonal covariance matrix, and the most extreme scenarios (those with the highest correlation between genes). The ellipsoids represent for each cell population the 95% confidence region and the red spherical icon and the green triangular icon represent respectively the centroids (average expression of gene 1 and gene 2) of cell population 1 and cell population 2.

structure (skeleton of the graph, namely identfication of the edges's coverage), represented as a binary adjacency matrix. Then, shift the binary network into a weighted structure, with different interaction weights between genes (as available in a precision matrix), normalised between -1 and 1 -¿ each cell population exhibits its own distribution profile, parametrised by the simulated network structure. Finally, reflect the heteroeignity of cell composition in biological tissues by drawing cellular proportions from a Dirichlet distribution.

Network generation

Un vaste pan de la littérature explored the generation of random and synthetic networks, each lying with its own assumptions. The most popular approaches are the Erdös-Rényi model (no particular structure), the preferential attachment model wich assumes a scale-free distribution of the degree of the network, and affiliation model, in which local structures prevail (see details in A). However, all these models return a binary adjacency matrix, encoding only for the presence or absence of an interaction.

The advantage of precision matrices precisely lies in the possibility of weighting edges by the strength of interaction, a higher absolute value depicting a stronger regulatory effect. Besides, the precision matrix should have the same sparsity level as the adjacency matrix, while being positive-definite. Inspiring from [7] and **huge**, we enforce these two constraints with 1:

$$\tilde{\boldsymbol{\Omega}} = \boldsymbol{A} \times v, \quad \boldsymbol{\Omega} = \tilde{\boldsymbol{\Omega}} + \mathrm{Diag}\left(\left|\min\left(\mathrm{Spectrum}\left(\tilde{\boldsymbol{\Omega}}\right)\right)\right| + u\right) \tag{1}$$

with u and v two scalar hyperparameters controlling the degree of connectivity in the graph. Precisely, v controls the magnitude of the interactions, with higher v inducing larger off-diagonal terms, while the second term of the equation $\mathrm{Diag}\left(\min\left(|\,\mathrm{Spectrum}\left(\tilde{\boldsymbol{\Omega}}\right)|\right) + u\right)$ guarantees the strict positiveness of the matrix (see 1). Accordingly, a fine balance must be found between the two hyperparameters, with a smaller v implying a more diagonal and well-conditioned matrix, while a larger v implies a stronger domination of the off-diagonal, interaction terms, and a larger spreading out of the spectrum.

add visual with eigenvalues plots to demonstrate affine invariance of the spectrum

Of note, we would have considered other simualtion framework sfor generating interesting Gaussian distirbution scenarios. For instance, the Toeplitz sructure, modelling an exponetial decay with distance beween variables, is commonly used for generating structured covariance matricees. Indeed, it"s controlled by a single paramter ρ, making it highly interpretable, and easy to compute with the Cholesky decomposition. However, since its primary focus is on the covariance, the framework can only be used to encode local dependencies (and not global, accounting for all higher-order interactions, as for the precision matrix) [8, 9].

The **mixSim** package, developed by [10], and the **clusterGeneration**, developed by [11], are other interesting alternatives for generating random clusters of cell populations, each modelled by a multivariate Gaussian distributions. However, **mixSim** plays on the global level of overlap between cell-type specific distributions, while **clusterGeneration** twists the level of degree separation between clusters, as quantified by a variant of the adjusted Rand Index.

3.3 Evaluation of robustness for DeCovarT

Sensitivity analyses, when the underlying distribution of counts is not Gaussian

- Add Gaussian white noise as a white-error term
- Evaluate sensibility to other types of statistical distributions [12]
- Dropout rate, as in MuSiC [12]
- Incomplete single-cell references, with missing cell populations [12]
- Evaluate tolerance for different single-cell RNASeq protocols [12]

4 Methods

4.1 DeCovarT model set-up

In this section, we recall the paradigm underlying most deconvolution algorithms, relating bulk transcriptomic expression to cell type-specific gene expression. Indices are fixed throughout: gene $g = 1, \dots, G$; cell type $j = 1, \dots, J$; sample (subject) $i = 1, \dots, N$ (independence across samples). Column and row slices use the conventional dot notation ($\boldsymbol{\mu}_{\cdot j}$, $\boldsymbol{y}_{\cdot i}$). Analogously to MuSiC [12], the signature stores *cross-sample mean* profiles rather than a single draw of latent expression. Notations:

- $\boldsymbol{y} = (y_{gi}) \in \mathbb{R}_+^{G\times N}$: bulk expression; $\boldsymbol{y}_{\cdot i} \in \mathbb{R}_+^G$ is the profile of sample i.
- $\boldsymbol{\mu} = (\mu_{gj}) \in \mathbb{R}^{G\times J}$: mean signature; $\boldsymbol{\mu}_{\cdot j} \in \mathbb{R}^G$ is the mean profile of cell type j (averaged over reference samples / cells, as in MuSiC).
- $\boldsymbol{p} = (p_{ji}) \in]0,1[^{J\times N}$: unknown relative proportions; $\boldsymbol{p}_{\cdot i} \in]0,1[^J$ is the composition of sample i.
- $\boldsymbol{\Sigma}_j \in \mathbb{R}^{G\times G}$ and $\boldsymbol{\Theta}_j \equiv \boldsymbol{\Sigma}_j^{-1}$: covariance and precision of cell type j. Stacking over j yields third-order tensors of size $G \times G \times J$ (one $G \times G$ network per cell type).

As in most traditional deconvolution models, bulk expression is the linear combination of mean cell-type profiles weighted by proportions (2):

$$\boldsymbol{y} = \boldsymbol{\mu}\,\boldsymbol{p} \quad (\text{equivalently } \boldsymbol{y}_{\cdot i} = \boldsymbol{\mu}\,\boldsymbol{p}_{\cdot i} \text{ for each } i). \tag{2}$$

In addition, each composition lies on the unit simplex (3):

$$\begin{cases} \sum_{j=1}^{J} p_{ji} = 1 \\ \forall j \in \widetilde{J} \quad p_{ji} \geq 0 \end{cases} \qquad (i = 1, \ldots, N). \tag{3}$$

4.2 Model assumptions

In contrast to standard linear regression, DeCovarT relaxes *exogeneity* by treating latent cell-type expression profiles as random rather than fixed. The closest published method is DSection [2], which implements a *univariate* Gaussian convolution (independent genes, gene-specific variances). DeMix [13], DeMixT [14] and ISOpureR [15] likewise treat tumour or stromal profiles as latent rather than observed; Ogundijo and Wang [16] extend DSection by approximating the posterior with sequential Monte Carlo. DeCovarT generalises the DSection likelihood to a *multivariate* Gaussian convolution with a sparse cell-type-specific covariance $\boldsymbol{\Sigma}_j$.

RNA-Sieve [17] is also a supervised, likelihood-based deconvolution, but of a different generative model. Writing their mean matrix M, mixing weights α and bulk b for DeCovarT's $\boldsymbol{\mu}$, $\boldsymbol{p}$ and $\boldsymbol{y}$, they treat b as the sum of n independent cells and invoke a *gene-wise* central-limit Gaussian whose variance $s_g^2(M, \alpha, S)$ depends on α. They further model estimation error in M ($\varepsilon_{M,gk} \sim \mathcal{N}(0, s_{gk}/c_k)$), an errors-in-variables correction of the design rather than a convolution of J latent type profiles. Genes are taken as independent. DeCovarT instead convolves multivariate type profiles and treats $(\boldsymbol{\mu}, \{\boldsymbol{\Sigma}_j\})$ as plug-in known.

Latent profile $\boldsymbol{x}_{\cdot j} \in \mathbb{R}^G$ of cell type j is multivariate Gaussian (4):

$$\operatorname{Det}(2\pi\boldsymbol{\Sigma}_j)^{-\frac{1}{2}} \exp\Big(-\tfrac{1}{2}(\boldsymbol{x}_{\cdot j} - \boldsymbol{\mu}_{\cdot j})^\top \boldsymbol{\Theta}_j (\boldsymbol{x}_{\cdot j} - \boldsymbol{\mu}_{\cdot j})\Big), \tag{4}$$

with parameters:

- $\boldsymbol{\mu}_{\cdot j}$: mean purified transcriptomic profile of cell type j (column of $\boldsymbol{\mu}$);

- $\mathbf{\Sigma}_j$: positive-definite covariance (Appendix 2), recovered via its inverse with `gLasso` [18]. We set $\mathbf{\Theta}_j \equiv \mathbf{\Sigma}_j^{-1}$; after normalisation, off-diagonal entries are partial correlations. Null off-diagonals are treated as absent edges.

Plug-in estimates $\zeta_j = (\boldsymbol{\mu}_{\cdot j}, \mathbf{\Sigma}_j)$ yield known $\boldsymbol{\zeta} = (\boldsymbol{\mu}, \{\mathbf{\Sigma}_j\}_{j=1}^J)$ with $\boldsymbol{\mu} = (\boldsymbol{\mu}_{\cdot j})_{j\in\widetilde{J}}$. Conditionally on $(\boldsymbol{\zeta}, \boldsymbol{p}_{\cdot i})$, the bulk $\boldsymbol{y}_{\cdot i}$ is a Gaussian convolution (5):

$$\boldsymbol{y}_{\cdot i} \,|\, (\boldsymbol{\zeta}, \boldsymbol{p}_{\cdot i}) \sim \mathcal{N}_G\big(\boldsymbol{\mu}\,\boldsymbol{p}_{\cdot i},\, \textstyle\sum_{j=1}^J p_{ji}^2 \mathbf{\Sigma}_j\big) \tag{5}$$

The directed graphical models associated with this framework are shown in Fig. 2: panel **a** recalls the classical linear deconvolution template; panel **b** is the unconstrained DeCovarT generative model; panel **c** adds the regularised soft-max mapping $\boldsymbol{\psi}$ that enforces the unit-simplex constraint on $\boldsymbol{p}$. Classical OLS and Gauss–Markov reminders (Article length) are deferred to the package vignette `DeCovarT-reminders-OLS`.

In the next section, we provide an explicit formula of the log-likelihood of our probabilistic framework, its gradient and hessian, which in turn can be used to retrieve the MLE of our distribution.

4.3 Derivation of the log-likelihood

From 5, the conditional log-likelihood is readily computed and given by 6:

$$\ell_{\boldsymbol{y}|\boldsymbol{\zeta}}(\boldsymbol{p}) = C + \frac{1}{2}\log\left(\operatorname{Det}\left(\sum_{j=1}^J p_j^2 \mathbf{\Sigma}_j\right)^{-1}\right) - \frac{1}{2}(\boldsymbol{y}-\boldsymbol{\mu}\boldsymbol{p})^\top \left(\sum_{j=1}^J p_j^2 \mathbf{\Sigma}_j\right)^{-1} (\boldsymbol{y}-\boldsymbol{\mu}\boldsymbol{p}) \tag{6}$$

Both the log-determinant and the Mahalanobis residual carry the factor $\frac{1}{2}$ of the Gaussian log-density (5). Keeping that factor is what makes the score and Hessian derived below consistent with the expected Fisher information used for Wald inference in B.6: taking expectations of 8 under the generative model returns $\mathcal{I}(\boldsymbol{p})$ exactly.

4.4 First and second-order derivation of the unconstrained DeCovarT log-likelihood function

The stationary points of a function and notably maxima, are given by the roots (the values at which the function crosses the x-axis) of its gradient, in our context, the vector: $\nabla\ell : \mathbb{R}^J \to \mathbb{R}^J$ evaluated at point $\nabla\ell(\boldsymbol{p}) :]0,1[^J \to \mathbb{R}^J$. Since the computation is the same for any cell ratio p_j, we give an explicit formula for only one of them (7):

$$
\begin{aligned}
\frac{\partial \ell_{\boldsymbol{y}|\boldsymbol{\zeta}}(\boldsymbol{p})}{\partial p_j} &= \tfrac{1}{2} \tfrac{\partial \log(\mathrm{Det}(\boldsymbol{\Theta}))}{\partial p_j} - \tfrac{1}{2} \left[\tfrac{\partial (\boldsymbol{y}-\boldsymbol{\mu}\boldsymbol{p})^\top}{\partial p_j} \boldsymbol{\Theta} (\boldsymbol{y}-\boldsymbol{\mu}\boldsymbol{p}) + (\boldsymbol{y}-\boldsymbol{\mu}\boldsymbol{p})^\top \tfrac{\partial \boldsymbol{\Theta}}{\partial p_j} (\boldsymbol{y}-\boldsymbol{\mu}\boldsymbol{p}) + (\boldsymbol{y}-\boldsymbol{\mu}\boldsymbol{p})^\top \boldsymbol{\Theta} \tfrac{\partial (\boldsymbol{y}-\boldsymbol{\mu}\boldsymbol{p})}{\partial p_j} \right] \\
&= -\tfrac{1}{2} \operatorname{Tr}\left(\boldsymbol{\Theta} \tfrac{\partial \boldsymbol{\Sigma}}{\partial p_j} \right) - \tfrac{1}{2} \left[-\boldsymbol{\mu}_{.j}^\top \boldsymbol{\Theta} (\boldsymbol{y}-\boldsymbol{\mu}\boldsymbol{p}) - (\boldsymbol{y}-\boldsymbol{\mu}\boldsymbol{p})^\top \Theta \tfrac{\partial \Sigma}{\partial p_j} \Theta (\boldsymbol{y}-\boldsymbol{\mu}\boldsymbol{p}) - (\boldsymbol{y}-\boldsymbol{\mu}\boldsymbol{p})^\top \boldsymbol{\Theta} \boldsymbol{\mu}_{.j} \right] \\
&= -p_j \operatorname{Tr}(\boldsymbol{\Theta}\boldsymbol{\Sigma}_j) + (\boldsymbol{y}-\boldsymbol{\mu}\boldsymbol{p})^\top \boldsymbol{\Theta} \boldsymbol{\mu}_{.j} + p_j (\boldsymbol{y}-\boldsymbol{\mu}\boldsymbol{p})^\top \boldsymbol{\Theta} \Sigma_j \boldsymbol{\Theta} (\boldsymbol{y}-\boldsymbol{\mu}\boldsymbol{p})
\end{aligned} \tag{7}
$$

Since the solution to $\nabla \left(\ell_{\boldsymbol{y}|\boldsymbol{\zeta}}(\boldsymbol{p}) \right) = 0$ is not closed, we had to approximate the MLE using iterated numerical optimisation methods. Some of them, such as the Levenberg–Marquardt algorithm, require a second-order approximation of the function, which needs the computation of the Hessian matrix. Deriving once more 7 yields the Hessian matrix, $\mathbf{H} \in \mathcal{M}_{J \times J}$ is given by:

$$
\begin{aligned}
\mathbf{H}_{i,i} = \frac{\partial^2 \ell}{\partial^2 p_i} &= -\operatorname{Tr}(\boldsymbol{\Theta}\boldsymbol{\Sigma}_i) + 2p_i^2 \operatorname{Tr}\left((\boldsymbol{\Theta}\boldsymbol{\Sigma}_i)^2 \right) - 2p_i (\boldsymbol{y}-\boldsymbol{\mu}\boldsymbol{p})^\top \boldsymbol{\Theta}\boldsymbol{\Sigma}_i \boldsymbol{\Theta} \boldsymbol{\mu}_{.i} - \boldsymbol{\mu}_{.i}^\top \boldsymbol{\Theta} \boldsymbol{\mu}_{.i} - \\
&\quad 2p_i (\boldsymbol{y}-\boldsymbol{\mu}\boldsymbol{p})^\top \boldsymbol{\Theta}\boldsymbol{\Sigma}_i \boldsymbol{\Theta} \boldsymbol{\mu}_{.i} - (\boldsymbol{y}-\boldsymbol{\mu}\boldsymbol{p})^\top \boldsymbol{\Theta} \left(4p_i^2 \boldsymbol{\Sigma}_i \boldsymbol{\Theta} \boldsymbol{\Sigma}_i - \boldsymbol{\Sigma}_i \right) \boldsymbol{\Theta} (\boldsymbol{y}-\boldsymbol{\mu}\boldsymbol{p}), \quad i \in \widetilde{J} \\
\mathbf{H}_{i,j} = \frac{\partial^2 \ell}{\partial p_i \partial p_j} &= 2p_j p_i \operatorname{Tr}(\boldsymbol{\Theta}\boldsymbol{\Sigma}_j \boldsymbol{\Theta}\boldsymbol{\Sigma}_i) - 2p_i (\boldsymbol{y}-\boldsymbol{\mu}\boldsymbol{p})^\top \boldsymbol{\Theta}\boldsymbol{\Sigma}_i \boldsymbol{\Theta} \boldsymbol{\mu}_{.j} - \boldsymbol{\mu}_{.i}^\top \boldsymbol{\Theta} \boldsymbol{\mu}_{.j} - \\
&\quad 2p_j (\boldsymbol{y}-\boldsymbol{\mu}\boldsymbol{p})^\top \boldsymbol{\Theta}\boldsymbol{\Sigma}_j \boldsymbol{\Theta} \boldsymbol{\mu}_{.i} - 4p_i p_j (\boldsymbol{y}-\boldsymbol{\mu}\boldsymbol{p})^\top \boldsymbol{\Theta}\boldsymbol{\Sigma}_i \boldsymbol{\Theta}\boldsymbol{\Sigma}_j \boldsymbol{\Theta} (\boldsymbol{y}-\boldsymbol{\mu}\boldsymbol{p}), \quad (i,j) \in \widetilde{J}^2, i \neq j
\end{aligned} \tag{8}
$$

in which the coloured sections pair one by one with the corresponding coloured sections of the gradient, given in 7.

Matrix calculus can largely ease the derivation of complex algebraic expressions, thus we remind in Appendices A.1 and A.2 relevant matrix properties and derivations. The numerical consistency of these derivatives was asserted with the **numDeriv** package, using the more stable Richardson's extrapolation

However, the explicit formulas for the gradient and the hessian matrix of the log-likelihood function, given in 7 and 8 respectively, do not take into account the simplex constraint assigned to the ratios. While some optimisation methods use heuristic methods to solve this problem, we consider alternatively a reparametrised version of the problem, detailed comprehensively in Appendix A.3.

5 Discussion

While other approaches, such as `DeMix` and `DeMixT`, enabled, using a sound statistical framework, to jointly estimate cellular portions and distributions of gene expression per sample and cell population, `DeCovarT` is the first deconvolution algorithm which explicitly handles the unit-simplex constraint, addressing the intrinsic compositional nature of cellular composition, while integrating interactions between genes.

Through extensive numerical simulations, playing on varying levels of gene

Hence, it provides a strong basis to further derive statistical tests to assert whether the proportion of a given cell population differs significantly between two distinct biological conditions.

5.1 Limitations and outlook

The present estimator is a closed-reference Gaussian convolution: every abundant cell type is assumed to appear as a column of $\boldsymbol{\mu}$, bulk and reference profiles are taken to lie on a compatible expression scale, and sample-level covariates, sequencing exposure, cell-type transcriptome size and condition-dependent shifts in $\boldsymbol{\mu}_{\cdot j}$ are not modelled. Graphical-lasso estimates of $\boldsymbol{\Theta}_j$ additionally shrink non-zero partial correlations. These restrictions motivate the extensions below; a longer catalogue with equations is given in the package perspectives vignette.

DeCovarT is not ordinary least squares. Cell-type covariances enter the residual law $\boldsymbol{\Sigma}(\boldsymbol{p}_{\cdot i})$ and cannot be treated as extra columns of $\boldsymbol{\mu}$. There is therefore no formula / `lm` interface, and no `predict()` method for forecasting bulk transcriptomes. Gene-wise affine standardisation (z-scoring $\boldsymbol{y}_{\cdot i}$, $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}_j$ with the same centre and scale) is equivariant for $\hat{\boldsymbol{p}}$ (2); logarithms and per-cell-type or per-sample transforms are not. Wald intervals use the expected Fisher information of the convolution mapped through the ALR (3). A systematic study of CPM / log2 normalisation, optimiser tolerance, and scaling with G, J and signature overlap is left to a later benchmark.

RNA-Sieve reports Wald regions from the inverse Fisher information of its gene-wise CLT likelihood, and a Godambe sandwich when protocol shift takes the truth outside the model family [17]. DeCovarT's analogue of the first is (3); it is not calibrated on a simplex face, where the chi-bar-square likelihood-ratio test and a restricted parametric bootstrap are required. A Godambe correction would be the natural sandwich if the multivariate convolution itself is misspecified. RNA-Sieve's noisy-M term is the likelihood analogue of resampling the reference; DeCovarT currently keeps $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}_j$ plug-in and propagates that uncertainty by a donor- or cell-level bootstrap. The package vignette on MLE properties compares the two generative models in notation aligned with (2).

Real-data evaluation should start from matched flow-cytometry / bulk RNA-seq panels such as the Kassandra benchmark [19], which already compares EPIC [20], CIBERSORT [1], CIBERSORTx [21], quanTIseq [22] and ABIS [23].

Observation laws and sample-level CTS.

DeCovarT is a continuous, frequentist convolution with plug-in $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}_j$. Discrete alternatives include multinomial–Dirichlet purification (ISOpureR, BayesPrism) [15, 24] and Poisson / Poisson–log-normal count models [25]. BLADE replaces the Gaussian by a gene-wise log-normal convolution and jointly infers cell-type-specific (CTS) expression [26]. A Bayesian CTS step for DeCovarT would recover $\boldsymbol{x}_{\cdot j,i}$ given $\boldsymbol{y}_{\cdot i}$ and $\boldsymbol{p}_{\cdot i}$ under the existing multivariate likelihood—the univariate analogue is DSection [2]. Isoform-level signatures $\boldsymbol{\mu}^{\text{iso}} \in \mathbb{R}^{T\times J}$ can further exploit differential transcript usage [27].

Covariates, unknown types, and RNA–cell uncoupling.

Sample-level covariates $\boldsymbol{z}_i$ cannot be appended as extra columns of $\boldsymbol{\mu}$. They may modulate cell-state moments $\boldsymbol{\mu}_j(\boldsymbol{z}_i)$ (as in MuSiC2, which drops condition-specific genes while remaining closed-reference) [28] or the ALR coordinates of $\boldsymbol{p}_{\cdot i}$. Incomplete references require an explicit unknown component $p_{u,i}\boldsymbol{u}_i$ or a signature adjustment:

DICEPro wraps existing engines when columns of $\boldsymbol{\mu}$ are missing [29]; BayICE adds a Bayesian unknown type with spike-and-slab selection [30]; EPIC already includes an uncharacterised compartment [20]. Independently, RNA mass fractions differ from cytometric cell fractions whenever transcriptome sizes r_j are heterogeneous. EPIC and quanTIseq apply a post-correction $\hat{p}_j^* \propto \hat{p}_j / r_j$ with measured or proxied r_j [20, 22, 31]; MMAD instead estimates r_j as an extra parameter, so the mixture is no longer linear [32].

Weighted and generalised least squares.

DWLS down-weights noisy genes by a diagonal W [33]. Ingesting those weights into BayesPrism is currently a strong joint $\boldsymbol{p}$+CTS combination [34]; the DeCovarT analogue is a weighted convolution $\boldsymbol{y}^\star = W^{1/2}\boldsymbol{y}$, $\boldsymbol{\Sigma}_j^\star = W^{1/2}\boldsymbol{\Sigma}_j W^{1/2}$ (2). Generalised least squares with a *global* residual covariance $\bar{\Sigma} = J^{-2}\sum_j \boldsymbol{\Sigma}_j$ (equi-balanced $p_j = 1/J$) is a simpler competitor that ignores cell-type-specific networks; cellGeometry estimates such a gene-level covariance from cross-sample variances [35]. Lineage trees add parent–child constraints on $\boldsymbol{p}$ [12]; elastic-net methods such as DCQ target which types *change* over time rather than a static simplex snapshot [36]; EnsDeconv fuses several bulk engines into one $\hat{\boldsymbol{p}}$ [37]. Each spot of a spatial assay is the same mixture with its own $\boldsymbol{p}(s)$ [38].

6 Code availability

All analyses were performed on ... high-performance computing infrastructure using StdEnv/2020. Core statistical analyses and deconvolution bench-marking were conducted in R (version 4.3.1), employing the following key packages: Seurat (v 4.3.0) for single-cell data processing; limma (v3.58.1), edgeR (v4.0.16), and DESeq2 (v1.42.1) for differential expression and discordant-gene filtering; lme4 (v1.1.35.5) and lmerTest (v3.1.3) for mixed-effects modelling; and tidyverse (v2.0.0) for data manipulation. Fifteen deconvolution methods were benchmarked, including EPIC (v1.1.7), FARDEEP (v1.0.1), MuSiC (v1.0.0), MuSiC2 (v0.1.0), BisqueRNA (v1.0.5), BayesPrism (v2.2.2), DWLS, hspe (v0.1), TOAST (v1.16.0), CSC-DRNA (v1.0.3), CSsingle (v0.1.0), DeconRNASeq (v1.44.0), and PREDE (v1.2.1). CIBERSORTx deconvolution was performed via its web interface (https://cibersortx.stanford.edu/). Additional methods, AutoGeneS and GEDIT, were implemented in Python (version 3.10) with numpy, pandas, and scipy.

Visualisation was performed using ggplot2 (v3.5.1), patchwork (v1.2.0), and cowplot (v1.1.3). To ensure reproducibility, all random seeds were fixed.

The package used to generate the simulations and infer ratios from virtual or real biological mixtures with the DeCovarT algorithm is implemented on Github DeCovarT.

7 Author contributions

B.C. conceived the method, implemented the `DeCovarT` package, designed the simulation studies, and wrote the manuscript. M.P. contributed equally to the methodological development and manuscript drafting. M.A.C. contributed to the benchmarking of gene-regulatory-network inference techniques used in the synthetic scenarios. G.N., E.B. and A.B. jointly supervised the work and revised the manuscript. All authors approved the final version.

8 Conflicts of interest

The authors declare no conflict of interest.

9 Acknowledgments

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Appendix A Synthetic network simulation

The three most popular approaches for generating random, synthetic networks are:

- The Erdős–Rényi model was first described in [39]: edges are generated independently between each pair of nodes with a fixed probability, leading to a homogeneous random graph.
- In the preferential attachment model ([40, 41]), nodes are iteratively added, favouring connection to the most well-connected nodes. This kind of framewrok is compliant with the scale-free network property, often looked for in biological systems, and resulting in the emergence of hubs. The degree distribution is notably given by a power-law.
- The affiliation model, also known as the stochastic block model (see [42]), assumes a latent indicator variable assigning each node to a community. The generation of edges is then dependent on group membership, with sronger interactions within a given block.

add a drawing of these three approaches

Theorem 1 (Invariance of the spectrum function by affine transformation) *Let X be any square matrix, and consider the affine transformation:*

$$Y = aX + bI$$

with a and b positive scalars, and I the identity matrix. Then, the eigenvectors of $\boldsymbol{Y}$ are the same as $\boldsymbol{X}$, and the eigenvalues are linearly transformed, shifted by b and scaled by a (larger range if a is above 1).

Appendix B Optimisation and calculus

B.1 Multivariate Gaussian reminders

Definition 1 (Multivariate Gaussian distributions) If random vector $\boldsymbol{X}$ of size G follows a random multivariate Gaussian distribution, $\boldsymbol{X} \sim \mathcal{N}_G(\mu, \boldsymbol{\Sigma})$, then its distribution is given by:

$$\mathrm{Det}(2\pi\Sigma)^{-\frac{1}{2}} \exp\left(-\frac{1}{2}(\boldsymbol{x}-\mu)\Sigma^{-1}(\boldsymbol{x}-\mu)^\top\right)$$

in which:

- $\mu = \boldsymbol{X}$ is the G-dimensional mean vector
- $\boldsymbol{\Sigma}$ is a $G \times G$ positive-definite 2 covariance matrix, whose diagonal terms, $\mathrm{Diag}(\boldsymbol{\Sigma}) = [(\mathbb{V}\mathrm{ar}\left[X_{i,j}\right]), \forall (i,j) \in \widetilde{G}^2, i = j]^\top$ are the individual variances of each purified gene transcript in population j and off-diagonal terms, $\boldsymbol{\Sigma}_{i,j} = \mathbb{C}\mathrm{ov}\left[X_i, X_j\right], \forall (i,j) \in \widetilde{G}^2, i \neq j$ are the covariance between variables. We note $\Theta = \Sigma^{-1}$, the inverse of the covariance matrix, called the *precision matrix*.

Remark 1 (Affine invariance property of multivariate GMMs) The two following properties hold for a multivariate Gaussian distribution:

- if $\boldsymbol{X} \sim \mathcal{N}_G(\mu, \boldsymbol{\Sigma})$, then $p\boldsymbol{X}$, with p a constant, follows itself a multivariate Gaussian distribution, given by: $p\boldsymbol{X} \sim \mathcal{N}_G(p\mu, p^2\Sigma)$
- given two independent random vectors $\boldsymbol{X_1} \sim \mathcal{N}_G(\mu_1, \boldsymbol{\Sigma_1})$ and $\boldsymbol{X_2} \sim \mathcal{N}_G(\mu_2, \boldsymbol{\Sigma_2})$ following a multivariate Gaussian distribution, then the random variable $\boldsymbol{X_1} + \boldsymbol{X_2}$ follows itself the multivariate Gaussian distribution:

  $$X + Y \sim \mathcal{N}_G(\mu_1 + \mu_2, \boldsymbol{\Sigma_1} + \boldsymbol{\Sigma_2})$$

  By induction, this property generalises to the sum of J independent random vectors of same dimension $\mathbb{R}^G$.

Remark 2 (Equivariance of the DeCovarT MLE) DeCovarT is equivariant to any fixed, invertible, gene-wise affine transformation applied consistently to bulk expression, reference means and reference covariances. Such transformations therefore do not alter the theoretical proportion estimates, although they may affect numerical conditioning. Transformations estimated separately for each cell type or bulk sample are not equivalent and are not supported.

Let $D = \mathrm{diag}(s_1, \ldots, s_G)$ with $s_g \neq 0$ and $\boldsymbol{m} \in \mathbb{R}^G$. The gene-wise map

$$\boldsymbol{y}^\star_{\cdot i} = D^{-1}(\boldsymbol{y}_{\cdot i} - \boldsymbol{m}), \quad \boldsymbol{\mu}^\star = D^{-1}(\boldsymbol{\mu} - \boldsymbol{m}\mathbf{1}_J^\top), \quad \boldsymbol{\Sigma}^\star_j = D^{-1}\boldsymbol{\Sigma}_j D^{-1}$$

preserves the linear convolution: because $\mathbf{1}_J^\top \boldsymbol{p}_{\cdot i} = 1$,

$$\boldsymbol{y}^\star_{\cdot i} = \boldsymbol{\mu}^\star \boldsymbol{p}_{\cdot i} + \boldsymbol{\varepsilon}^\star_{\cdot i}$$

with the same $\boldsymbol{p}_{\cdot i}$. The Mahalanobis residual is invariant,

$$(\boldsymbol{y} - \boldsymbol{\mu}\boldsymbol{p})^\top \boldsymbol{\Sigma}(\boldsymbol{p})^{-1}(\boldsymbol{y} - \boldsymbol{\mu}\boldsymbol{p}) = (\boldsymbol{y}^\star - \boldsymbol{\mu}^\star\boldsymbol{p})^\top \boldsymbol{\Sigma}^\star(\boldsymbol{p})^{-1}(\boldsymbol{y}^\star - \boldsymbol{\mu}^\star\boldsymbol{p}),$$

so the maximiser $\hat{\boldsymbol{p}}$ is unchanged. A z-score (centre and scale computed once per gene from $\boldsymbol{\mu}$, including any gene-wise covariate adjustment of those means) is the special case $m_g = \mathrm{mean}_g$, $s_g = \mathrm{sd}_g$.

A nonlinear logarithm is a different matter. CIBERSORT requires non-negative expression, no missing values, and a non-log linear scale [1]. By Jensen's inequality the concave map log changes first and second moments, and $\log(\boldsymbol{\mu}\boldsymbol{p}) \neq (\log \boldsymbol{\mu})\boldsymbol{p}$ except at degenerate points, so the convolution is no longer linear. The logarithm is continuous and strictly increasing on $(0, \infty)$, hence order-preserving for strictly positive expression, but that does *not* leave the Gaussian-convolution MLE of $\boldsymbol{p}$ invariant.

Remark 3 (Expected Fisher information and Wald intervals) For one bulk sample the convolution is $\boldsymbol{y}_{\cdot i} \sim \mathcal{N}_G(\boldsymbol{\mu}\boldsymbol{p}, \boldsymbol{\Sigma}(\boldsymbol{p}))$ with $\boldsymbol{\Sigma}(\boldsymbol{p}) = \sum_j p_j^2 \boldsymbol{\Sigma}_j$. Writing $\boldsymbol{\Theta}(\boldsymbol{p}) = \boldsymbol{\Sigma}(\boldsymbol{p})^{-1}$, the expected Fisher information of the unconstrained mean–covariance map is

$$I(\boldsymbol{p})_{jk} = \boldsymbol{\mu}_{\cdot j}^\top \boldsymbol{\Theta}(\boldsymbol{p})\boldsymbol{\mu}_{\cdot k} + 2p_j p_k \operatorname{tr}\big(\boldsymbol{\Theta}(\boldsymbol{p})\boldsymbol{\Sigma}_j \boldsymbol{\Theta}(\boldsymbol{p})\boldsymbol{\Sigma}_k\big). \tag{B1}$$

Cramér–Rao supplies the lower bound $\operatorname{Var}(\hat{\boldsymbol{\rho}}) \succeq I_{\boldsymbol{\rho}}^{-1}$ in ALR coordinates $\boldsymbol{\rho}$, where $I_{\boldsymbol{\rho}} = \mathbf{J}_{\boldsymbol{\psi}}^\top I(\boldsymbol{p})\mathbf{J}_{\boldsymbol{\psi}}$ and $\mathbf{J}_{\boldsymbol{\psi}} = \partial\psi/\partial\boldsymbol{\rho}^\top$. The delta method maps that bound back to the simplex:

$$\operatorname{Var}(\hat{\boldsymbol{p}}) = \mathbf{J}_{\boldsymbol{\psi}} I_{\boldsymbol{\rho}}^{-1} \mathbf{J}_{\boldsymbol{\psi}}^\top. \tag{B2}$$

This is the sandwich returned by `vcov()` on a fitted DeCovarT object. It is not an OLS residual variance: goodness of fit is the convolution log-likelihood, and the estimator is not a linear model for forecasting bulk expression.

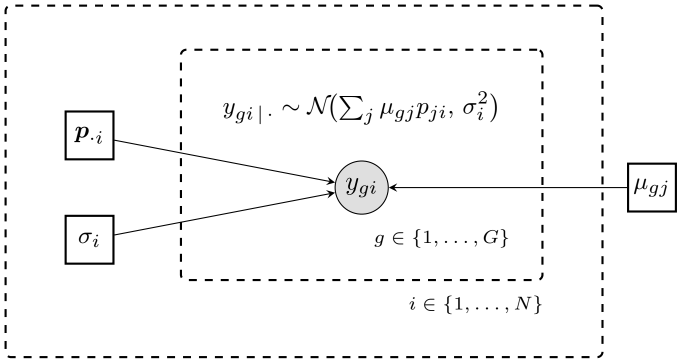
Deriving the characteristic function of the multivariate GMM yields directly results reported in 1.

Definition 2 (Definite matrix) A symmetric real matrix $\boldsymbol{A}$ of rank G is *positive-definite* if:

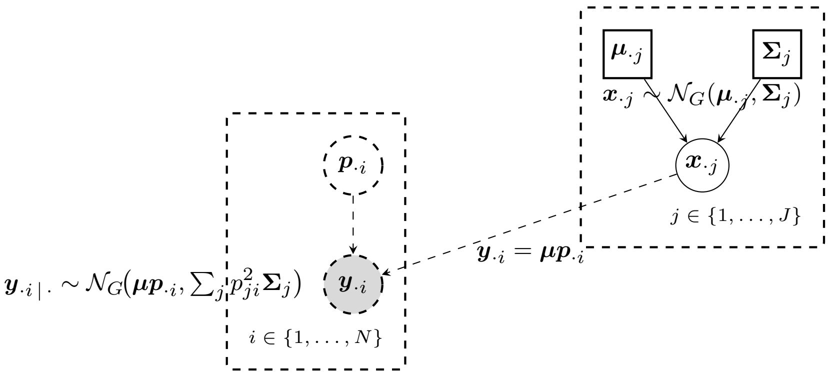
$$\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x} > 0, \quad \boldsymbol{x} \in \mathbb{R}^G \tag{B3}$$

To gain a clearer grasp of the positive-definite constraint imposed on the covariance parameter of a multivariate Gaussian distribution, let's delve into the most straightforward scenario, in which we assume that any of the individual features exhibit pairwise independence. This particular setup is parametrised by a covariance matrix containing exclusively diagonal elements.

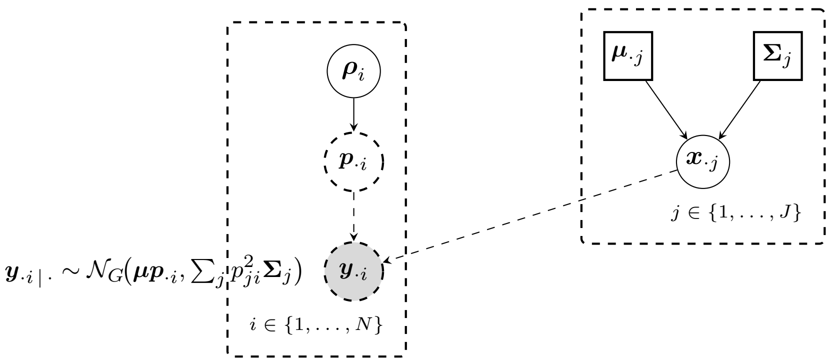
If the matrix is not strictly positive-definite, then some of the diagonal elements can display negative values, otherwise that the individual variances for some of the covariates are negative. It is not physically possible and leads to improper, degenerate probability distributions.



a Standard linear deconvolution (gold-standard template).



b Unconstrained DeCovarT generative model.



c Constrained DeCovarT (regularised soft-max on the simplex).

Fig. 2 Directed graphical models (plate notation), following the conventions of RevBayes. Shaded nodes are observed; square nodes are fixed (plug-in) parameters; dashed circles denote deterministic or constrained quantities. **a** Gold-standard linear model $y_{gi\,|\,\cdot} \sim \mathcal{N}(\sum_j \mu_{gj} p_{ji}, \sigma_i^2)$. **b** Unconstrained DeCovarT convolution $\boldsymbol{y}_{\cdot i\,|\,\cdot} \sim \mathcal{N}_G(\boldsymbol{\mu}\boldsymbol{p}_{\cdot i}, \sum_j p_{ji}^2 \boldsymbol{\Sigma}_j)$ with $\boldsymbol{x}_{\cdot j} \sim \mathcal{N}_G(\boldsymbol{\mu}_{\cdot j}, \boldsymbol{\Sigma}_j)$. **c** Constrained DeCovarT: unconstrained coordinates $\boldsymbol{\rho}_i \in \mathbb{R}^{J-1}$ are mapped to the open simplex by the C^2-diffeomorphism $\boldsymbol{\psi} : \boldsymbol{\rho} \mapsto \boldsymbol{p}$ with $p_j = e^{\rho_j}/(\sum_{k<J} e^{\rho_k} + 1)$ for $j < J$, $p_J = 1/(\sum_{k<J} e^{\rho_k} + 1)$, and inverse $\boldsymbol{\psi}^{-1}(\boldsymbol{p}) = \big(\ln(p_j/p_J)\big)_{j<J}$ (Eq. B4). Variability is attributed to latent covariates $\boldsymbol{x}_{\cdot j}$.

B.2 Matrix calculus

Fundamental algebra calculus formulas used to derive first-order and second-order derivatives of the generative model of `DeCovarT` are reported in 4 and 5, respectively.

Remark 4 (First-order matrix calculus) Given two invertible matrices, $A = \boldsymbol{A}(p)$ and $B = \boldsymbol{B}(p)$, functions of a scalar variable p, the following matrix calculus hold:

(a) $\frac{\partial \operatorname{Det}(\boldsymbol{A})}{\partial p} = \operatorname{Det}(\boldsymbol{A}) \operatorname{Tr}\left(\boldsymbol{A}^{-1} \frac{\partial \boldsymbol{A}}{\partial p}\right)$ (b) $\frac{\partial \boldsymbol{U}\boldsymbol{A}\boldsymbol{V}}{\partial p} = \boldsymbol{U} \frac{\partial \boldsymbol{A}}{\partial p} \boldsymbol{V}$ (c) $\frac{\partial \boldsymbol{A}^{-1}}{\partial p} = -\boldsymbol{A}^{-1} \frac{\partial \boldsymbol{A}}{\partial p} \boldsymbol{A}^{-1}$

From a) and fundamental linear algebra properties, we can readily compute applying the chain rule property on the logarithm:

$$\frac{\partial \log\left(\operatorname{Det}(\boldsymbol{A})\right)}{\partial p} = \operatorname{Tr}\left(\boldsymbol{A}^{-1} \frac{\partial \boldsymbol{A}}{\partial p}\right)$$
$$\frac{\partial \log\left(\operatorname{Det}(\boldsymbol{A}^{-1})\right)}{\partial p} = -\operatorname{Tr}\left(\boldsymbol{A}^{-1} \frac{\partial \boldsymbol{A}}{\partial p}\right)$$

Finally, injecting these first-order matrix derivatives, we obtain:

$$\begin{aligned}\frac{\partial (\boldsymbol{y} - \boldsymbol{x}p)^\top \Theta (\boldsymbol{y} - \boldsymbol{x}p)}{\partial p} &= -2(\boldsymbol{y} - \boldsymbol{x}p)^\top \Theta \boldsymbol{x} \\ &= -2\boldsymbol{x}^\top \Theta (\boldsymbol{y} - \boldsymbol{x}p)\end{aligned}$$
$$\text{with } \boldsymbol{A} = \boldsymbol{D} = -\boldsymbol{x} \in \mathbb{R}^G, \quad \boldsymbol{b} = \boldsymbol{e} = \boldsymbol{y}, \quad \boldsymbol{C} = \boldsymbol{\Theta} \text{ symmetric}$$

Remark 5 (Second-order matrix calculus) Given an invertible matrix $\boldsymbol{A}$ depending on a variable p, the following calculus formulas hold:

(a) $\frac{\partial^2 \boldsymbol{A}^{-1}}{\partial p_i \partial p_j} = \boldsymbol{A}^{-1} \left(\frac{\partial \boldsymbol{A}}{\partial p_i} \boldsymbol{A}^{-1} \frac{\partial \boldsymbol{A}}{\partial p_j} - \frac{\partial^2 \boldsymbol{A}}{\partial p_i \partial p_j} + \frac{\partial \boldsymbol{A}}{\partial p_j} \boldsymbol{A}^{-1} \frac{\partial \boldsymbol{A}}{\partial p_i}\right) \boldsymbol{A}^{-1}$ (b) $\frac{\partial \operatorname{Tr}(\boldsymbol{A})}{\partial p_i} = \operatorname{Tr}\left(\frac{\partial \boldsymbol{A}}{\partial p_i}\right)$

Combining 4 with the linear property of the trace operator yields:

$$\frac{\partial^2 \log\left(\operatorname{Det}(\boldsymbol{A}^{-1})\right)}{\partial^2 p} = -\operatorname{Tr}\left[\boldsymbol{A}^{-1} \frac{\partial^2 \boldsymbol{A}}{\partial^2 p_i}\right] + \operatorname{Tr}\left[\left(\boldsymbol{A}^{-1} \frac{\partial \boldsymbol{A}}{\partial p_i}\right)^2\right]$$

B.3 First and second-order derivation of constrained DeCovarT

B.4 Review of constrained optimisation approaches

Constrained optimization can be addressed either by eliminating constraints through re-parametrisation, softly enforcing them via penalties or barriers, or exactly enforcing feasibility using projections, Lagrangian, or KKT-based methods. Re-parametrisation and penalties are simple but poorly conditioned; barrier and interior-point methods are powerful for convex problems; proximal and projection methods excel in large-scale structured settings; and Lagrangian-based approaches (ALM, SQP, MMA) provide the most general and theoretically grounded framework for high-accuracy solutions.

To reparametrise the log-likelihood function (6) in order to explicitly handling the unit simplex constraint (3), we consider the following mapping function: $\boldsymbol{\psi} : \boldsymbol{\rho} \rightarrow \boldsymbol{p} \mid \quad \boldsymbol{\rho} \in \mathbb{R}^{J-1}, \boldsymbol{p} \in]0,1[^J$ (B4):

1. $$\boldsymbol{p} = \boldsymbol{\psi}(\boldsymbol{\rho}) = \begin{cases} p_j = \frac{e^{\rho_j}}{\sum_{k<J} e^{\rho_k} + 1}, \, j < J \\ p_J = \frac{1}{\sum_{k<J} e^{\rho_j} + 1} \end{cases} \tag{B4}$$

2. $\boldsymbol{\rho} = \boldsymbol{\psi}^{-1}(\boldsymbol{p}) = \left(\ln\left(\frac{p_j}{p_J}\right)\right)_{j \in \{1,\dots,J-1\}}$

that is a C^2-diffeomorphism, since $\boldsymbol{\psi}$ is a bijection between $\boldsymbol{p}$ and $\boldsymbol{\rho}$ twice differentiable.

Its Jacobian, $\mathbf{J}_{\boldsymbol{\psi}} \in \mathcal{M}_{J \times (J-1)}$ is given by B5:

$$\mathbf{J}_{i,j} = \frac{\partial p_i}{\partial \rho_j} = \begin{cases} \frac{e^{\rho_i} B_i}{A^2}, \quad i = j, \, i < J \\ \frac{-e^{\rho_j} e^{\rho_i}}{A^2}, \quad i \neq j, \, i < J \\ \frac{-e^{\rho_j}}{A^2}, \quad i = J \end{cases} \tag{B5}$$

with i indexing vector-valued $\boldsymbol{p}$ and j indexing the first-order order partial derivatives of the mapping function, $A = \sum_{j'<J} e^{\rho_{j'}} + 1$ the sum over exponential (denominator of the mapping function) and $B = A - e^{\rho_i}$ the sum over ratios minus the exponential indexed with the currently considered index i.

The Hessian of the multi-dimensional mapping function $\boldsymbol{\psi}(\boldsymbol{\rho})$ exhibits symmetry for each cell ratio component j, as anticipated in accordance with Schwarz's theorem. It is is a third-order tensor of rank $(J-1)(J-1)J$, given by B6:

$$\frac{\partial^2 p_i}{\partial k \partial j} = \begin{cases} \frac{e^{\rho_i} e^{\rho_l}(-B_i + e^{\rho_i})}{A^3}, \, (i < J) \wedge ((i \neq j) \oplus (i \neq k)) \quad (a) \\ \frac{2 e^{\rho_i} e^{\rho_j} e^{\rho_k}}{A^3}, \, (i < J) \wedge (i \neq j \neq k) \quad (b) \\ \frac{e^{\rho_i} e^{\rho_j}(-A + 2e^{\rho_j})}{A^3}, \, (i < J) \wedge (j = k \neq i) \quad (c) \\ \frac{B_i e^{\rho_i}(B_i - e^{\rho_i})}{A^3}, \, (i < J) \wedge (j = k = i) \quad (d) \\ \frac{e^{\rho_j}(-A + 2e^{\rho_j})}{A^3}, \, (i = J) \wedge (j = k) \quad (e) \\ \frac{2 e^{\rho_j} e^{\rho_k}}{A^3}, \, (i = J) \wedge (j \neq k) \quad (f) \end{cases} \tag{B6}$$

with i indexing $\boldsymbol{p}$, j and k respectively indexing the first-order and second-order partial derivatives of the mapping function with respect to $\boldsymbol{\rho}$. In line (a), $\oplus$ refers to the Boolean XOR operator, $\wedge$ to the AND operator and $l = \{j, k\} \setminus i$.

To derive the log-likelihood function in 7, we reparametrise $\boldsymbol{p}$ to $\boldsymbol{\rho}$, using a standard *chain rule formula*. Considering the original log-likelihood function, 6, and the mapping function, B4, the differential at the first order and at the second order is given by B7 and B8, respectively defined in $\mathbb{R}^{J-1}$ and $\mathcal{M}_{(J-1)\times(J-1)}$:

$$\left[\frac{\partial \ell_{\boldsymbol{y}|\boldsymbol{\zeta}}}{\partial \rho_j}\right]_{j<J} = \sum_{i=1}^{J} \frac{\partial \ell_{\boldsymbol{y}|\boldsymbol{\zeta}}}{\partial p_i} \frac{\partial p_i}{\partial \rho_j} \tag{B7}$$

$$\left[\frac{\partial \ell^2_{\boldsymbol{y}|\boldsymbol{\zeta}}}{\partial \rho_k \rho_j}\right]_{j<J,\, k<J} = \sum_{i=1}^{J}\sum_{l=1}^{J} \left(\frac{\partial p_i}{\partial \rho_j} \frac{\partial^2 \ell_{\boldsymbol{y}|\boldsymbol{\zeta}}}{\partial p_i \partial p_l} \frac{\partial p_l}{\partial \rho_k}\right) + \sum_{i=1}^{J} \left(\frac{\partial \ell_{\boldsymbol{y}|\boldsymbol{\zeta}}}{\partial p_i} \frac{\partial^2 p_i}{\partial \rho_k \rho_j}\right) \quad (d) \tag{B8}$$

B.5 Numerical optimisation algorithms

The extremum, and by extension the MLE, is a root of the gradient of the log-likelihood. However, in our generative framework, the inverse function cancelling the gradient of 6 is non-closed. Instead, iterated numerical optimisation algorithms that consider first or second-order approximations of the function to optimise are used to approximate the roots.

The *Levenberg-Marquardt (LM)* algorithm [43] bridges the gap between between the steepest descent method (first-order) and the Newton-Raphson method (second-order) by inflating the diagonal terms of the Hessian matrix. Far from the endpoint, a second-order descent is favoured for its faster convergence pace, while the steepest approach is privileged close to the extremum since it allows careful refinement of the step size. Specially, we used the LM implementation of R package `marqLevAlg` to infer estimates of the cellular ratios from the bootstrap simulations ([44]). It notably includes an additional convergence criteria, the relative distance to the maximum (RDM), that sets apart extrema from spurious saddle points.

B.6 Asymptotic inference and boundary calibration

Expectations of 8 under the generative model, using $\mathbb{E}[\boldsymbol{r}] = \boldsymbol{0}$ and $\mathbb{E}[\boldsymbol{r}\boldsymbol{r}^\top] = \boldsymbol{\Sigma}(\boldsymbol{p})$ with $\boldsymbol{r} = \boldsymbol{y} - \boldsymbol{\mu}\boldsymbol{p}$, cancel the determinant traces against the residual traces and return exactly the expected Fisher information B1 of 3, that is $\mathbb{E}[-\mathbf{H}] = I(\boldsymbol{p})$. This identity holds only with the factor $\frac{1}{2}$ of 6 in place; doubling the log-determinant, as an earlier version of the implementation did, breaks it and leaves the objective inconsistent with the Wald standard errors obtained from B2 [45]. Two caveats bound the interpretation of those standard errors.

First, the realised log-likelihood is *not* globally concave: with $\boldsymbol{\Sigma}_1 = \boldsymbol{\Sigma}_2 = \mathbf{I}_3$, $\boldsymbol{\mu} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \\ 1 & 6/5 \end{pmatrix}$ and $\boldsymbol{y} = \boldsymbol{\mu}(1/2, 1/2)^\top$, the Hessian at the interior point $\boldsymbol{p} = (1/5, 3/10)$ has eigenvalues ≈ -654.5 and $+53.9$, hence is indefinite. A converged optimiser therefore certifies a local maximum only. Uniqueness is recovered at the *population* level, where the Kullback-Leibler identity makes $\boldsymbol{p}_0$ the unique maximiser of $\mathbb{E}_{\boldsymbol{p}_0}\{\ell(\boldsymbol{p})\}$ under the rank condition of 4.2, and the local curvature of that divergence is exactly **??** [46].

Second, testing the absence of a cell type, $H_0 : p_j = 0$ against $H_1 : p_j \geq 0$, places the null on the boundary of the simplex. The local parameter space is then a half-line rather than a vector space, so Wilks' theorem [47] does not apply and the additive log-ratio chart cannot even represent the null ($\rho_j = -\infty$). Following [48] and [49], the problem is asymptotically equivalent to projecting a Gaussian score onto the tangent cone approximating the admissible set, and with q active boundary constraints and s interior constraints the profile likelihood-ratio statistic $D = 2\{\ell(\hat{\boldsymbol{p}}) - \ell(\hat{\boldsymbol{p}}^{(0)})\}$ converges to the chi-bar-square mixture (B9):

$$D \rightsquigarrow \sum_{i=0}^{q} \binom{q}{i} 2^{-q} \chi^2_{s+i} \tag{B9}$$

For a single absent cell type B9 is the $\frac{1}{2}\chi^2_0 + \frac{1}{2}\chi^2_1$ law, so the correct one-sided p-value is half the naive one. The binomial weights require the active information block to be orthogonal to the nuisance block; we therefore also provide a restricted parametric bootstrap, which resimulates $\boldsymbol{Y}^{(b)} \sim \mathcal{N}_G(\boldsymbol{\mu p}, \boldsymbol{\Sigma}(\boldsymbol{p}))$ under the restricted MLE and recalibrates D without invoking B9. Because a single bulk profile provides no replication for its own composition, all of these limits are taken in the number of replicate samples sharing one composition.

Appendix C Special cases of the generative model

C.1 Simplifications adopting the marker-based paradigm

A simpler case to derive the log-likelihood is adopting a marker-based paradigm, as we can factorise the multivariate conditional distribution as the product of J conditional independent multivariate Gaussian distributions, each defined on a known gene marker set G_j and with only parameter to estimate p_j, formally C10:

$$\mathrm{P}_{\boldsymbol{p}}(\boldsymbol{Y}|\boldsymbol{X}, \Sigma) = \mathrm{P}i_{j=1}^{J} \mathrm{P}_{p_j}(\mathrm{y}_{G_j}|\mathrm{x}_{G_j}, \Sigma_j) \tag{C10}$$

The log-likelihood in a marker-based context is given by C11:

$$\begin{aligned} \ell(\boldsymbol{p}|\boldsymbol{y}, \boldsymbol{X}, \Sigma) &= \textstyle\sum_{j=1}^{J} \mathbb{P}_{p_j}(\mathrm{y}_{G_j}|\mathrm{x}_{G_j}, \Sigma_j) \\ &= C + \textstyle\sum_{j=1}^{J} \left[-|G_j| \log(p_j) - \frac{1}{2p_j^2}(\mathrm{y}_{G_j} - \mathrm{x}_{G_j} p_j)\Theta_j(\mathrm{y}_{G_j} - \mathrm{x}_{G_j} p_j)^\top \right] \text{ (a)} \\ &= C + \textstyle\sum_{j=1}^{J} \left[-|G_j| \log(p_j) - \frac{1}{2p_j^2} \sum_{i=g}^{G_j} \Theta_{gg}\epsilon_{gg}^2 - \frac{1}{p_j^2} \sum_{1 \leq g < l \leq |G_j|} \Theta_{gl}\epsilon_{gl}^2 \right] \text{ (b)} \end{aligned} \tag{C11}$$

with $C = \frac{1}{2}\sum_{j=1}^{J} \log(\det(\Theta_j)) + \frac{1}{2} GJ \log(2\pi)$ a constant, $\epsilon_{gg} = (y_g - p_j x_g)$ the residual of canonical linear regression of gene g in market set G_j, $\epsilon_{gl} = (y_g - p_j x_g)(y_l - p_j x_l)$, $g \neq l$ and Θ_{gl} input of the precision matrix indexed by row index g and column index l.

Now, computing the first-order derivative of the log-likelihood using matrix calculus 5 and setting to 0, we get C12:

$$\begin{aligned}
\frac{\partial \ell(\boldsymbol{p}|\boldsymbol{y},\boldsymbol{X},\Sigma)}{\partial p_j} &= 0 \\
&\iff \\
-\frac{|G_j|}{p_j} + \frac{(\mathrm{y}_{G_j}-\mathrm{x}_{G_j}p_j)^\top \Theta_j (\mathrm{y}_{G_j}-\mathrm{x}_{G_j}p_j)}{p_j^3} + \frac{x_{G_j}^\top \Theta_j (\mathrm{y}_{G_j}-\mathrm{x}_{G_j}p_j)}{p_j^2} &= 0 \\
&\iff \\
-|G_j| + \frac{\mathrm{y}_{G_j}^\top \Theta_j \mathrm{y}_{G_j}}{p_j^2} - 2\frac{\mathrm{x}_{G_j}^\top \Theta_j \mathrm{y}_{G_j}}{p_j} + \cancel{\mathrm{x}_{G_j}^\top \Theta_j \mathrm{x}_{G_j}} + \frac{x_{G_j}^\top \Theta_j \mathrm{y}_{G_j}}{p_j} - \cancel{\mathrm{x}_{G_j}^\top \Theta_j \mathrm{x}_{G_j}} &= 0 \\
&\iff \\
-|G_j| p_j^2 - \mathrm{x}_{G_j}^\top \Theta_j \mathrm{y}_{G_j} p_j + \mathrm{y}_{G_j}^\top \Theta_j \mathrm{y}_{G_j} &= 0
\end{aligned} \tag{C12}$$

Solving quadratic equation C12 yields two real roots corresponding to two extrema of the log-likelihood function:

$$(\hat{p_{j1}}, \hat{p_{j2}}) = \frac{-\boldsymbol{x}_{G_j}^\top \Theta_j \mathrm{y}_{G_j} \pm \sqrt{\left(\boldsymbol{x}_{G_j}^\top \Theta_j \mathrm{y}_{G_j}\right)^2 + 4|G_j| \mathrm{y}_{G_j}^\top \Theta_j \mathrm{y}_{G_j}}}{2|G_j|} \tag{C13}$$

Indeed, $\Delta = \left(\boldsymbol{x}_{G_j}^\top \Theta_j \mathrm{y}_{G_j}\right)^2 + 4|G_j| \mathrm{y}_{G_j}^\top \Theta_j \mathrm{y}_{G_j}$ is strictly positive, with first term of the discriminant squared raised and second term strictly positive from the condition of positiveness imposed on the covariance matrix 2.

To identify the MLE, we should check which of the two solutions is a maximum, by computing the second-order derivative of the log likelihood with respect to $\hat{p}$ and assert which one is negative. The second derivative is given by the following formula C14:

$$\frac{\partial^2 \ell(\boldsymbol{p}|\boldsymbol{y},\boldsymbol{X},\Sigma)}{\partial^2 p_j} = \frac{|G_j|}{p_j^2} - 3\frac{\mathrm{y}_{G_j}^\top \Theta_j \mathrm{y}_{G_j}}{p_j^4} + 2\frac{x_{G_j}^\top \Theta_j \mathrm{y}_{G_j}}{p_j^3} \tag{C14}$$

associated to the two following real roots C15:

$$\frac{-x_{G_j}^\top \Theta_j \mathrm{y}_{G_j} \pm \sqrt{\left(\boldsymbol{x}_{G_j}^\top \Theta_j \mathrm{y}_{G_j}\right)^2 + 3|G_j| \mathrm{y}_{G_j}^\top \Theta_j \mathrm{y}_{G_j}}}{|G_j|} \tag{C15}$$

Using the property that a quadratic function (we make it appeared by multiplying by the always positive factor p^4) has an opposite sign between its roots to that of its quadratic term (in C14, the always positive term $|G_j|$), we deduce that the MLE estimate, $\hat{p_j}$, as computed in C13 is indeed a maximum if it's included in the interval defined in C15.

C.2 Robust linear regression with GLS

DeCovarT's residual covariance $\Sigma(\boldsymbol{p}_{\cdot i}) = \sum_{j=1}^{J} p_{ji}^2 \boldsymbol{\Sigma}_j$ depends on the unknown composition. An ad-hoc GLS approximation freezes that map at equi-balanced weights

$p_j = 1/J$,

$$\mathrm{Cov}(\boldsymbol{y}_{\cdot i} \mid \boldsymbol{\mu}) = \frac{1}{J^2}\sum_{j=1}^{J}\boldsymbol{\Sigma}_j, \tag{C16}$$

so the variance of purified profiles is treated as independent of their mixing weights. This is exactly generalised least squares (GLS) on the linear mean $\boldsymbol{y}_{\cdot i} = \boldsymbol{\mu}\,\boldsymbol{p}_{\cdot i}$ 2: a *single* residual covariance Σ, not a cell-type tensor. cellGeometry estimates such a global gene-level covariance from cross-sample variances and gene-wise heteroscedasticity [35]. DeCovarT remains strictly richer whenever $\boldsymbol{\Sigma}_j$ differ across j and Σ should scale with p_{ji}^2.

Weighted least squares (WLS) is the diagonal special case $\Theta = \mathrm{diag}(w_1, \dots, w_G)$. DWLS supplies dampened gene weights w_g for that diagonal [33]. A direct DeCovarT extension, following the BayesPrism–DWLS construction [34], is to ingest $W = \mathrm{diag}(w_g)$ into the convolution without dropping cell-type-specific networks: set $\boldsymbol{y}^\star = W^{1/2}\boldsymbol{y}_{\cdot i}$, $\boldsymbol{\mu}^\star = W^{1/2}\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}_j^\star = W^{1/2}\boldsymbol{\Sigma}_j W^{1/2}$, then run the existing simplex-constrained solver. Interaction monomials $p_j p_k$ in the mean (a general linear model / Scheffé expansion) can be added on the same GLS residual, at the cost of many extra $\boldsymbol{\mu}_{jk}$ parameters.

Theorem 2 (Generalised Least Squares Regression) *Consider the multivariate linear model*

1. $\boldsymbol{y}_{\cdot i} = \boldsymbol{\mu}\,\boldsymbol{p}_{\cdot i} + \epsilon$
2. $\mathbb{E}(\epsilon \mid \boldsymbol{\mu}) = 0$
3. $\mathrm{Cov}(\epsilon \mid \boldsymbol{\mu}) = \Sigma$

with Σ a known non-singular covariance. The GLS estimate that minimises the squared Mahalanobis distance of the residuals is

$$\hat{\boldsymbol{p}}^{\mathrm{GLS}} = (\boldsymbol{\mu}^\top\Theta\boldsymbol{\mu})^{-1}\boldsymbol{\mu}^\top\Theta\boldsymbol{y}_{\cdot i},$$

where $\Theta = \Sigma^{-1}$. The estimator is unbiased, consistent, efficient and asymptotically normal; under a Gaussian conditional law with mean $\boldsymbol{\mu}\,\boldsymbol{p}_{\cdot i}$ and covariance Σ it coincides with the unconstrained MLE. A Cholesky factor $\Sigma = \boldsymbol{C}\boldsymbol{C}^\top$ reduces the problem to OLS after the transform $\boldsymbol{C}^{-1}\boldsymbol{y}_{\cdot i}$, removing heteroscedasticity and residual correlation. The GLS estimate is the best linear unbiased estimator of $\boldsymbol{p}_{\cdot i}$ in this unconstrained model; non-negativity and $\sum_j p_{ji} = 1$ are imposed afterwards, or via the ALR map used in the main text. [1]

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